Blackout

System Operator · Calder Interconnection

10 DAYS · 41 STOPS · GRADE 6 · BLACKOUT_MS

THE OPENING CARD

Four million people are supplied from the Calder network, and the whole of it runs on one number. Mains frequency has to stay within half a hertz of fifty. Drift further and generators start tripping off, which takes the rest of the system with them. You are the system operator on nights. The power stations belong to companies you can instruct under contract and cannot order about. Demand rises and falls with whatever four million people happen to be doing. Holding the two together, second by second, is the job. A fault takes seconds. Getting a system back after one has gone down takes days, and the hospitals on this network hold about eight hours of their own generation.

DAY 1 OF 10

The number on the wall

BEFORE THE DAY — GREET

The shift changes at seven and nobody has met you

Two crews work here and they swap over at seven. At four in the morning a voice on the radio is either somebody you have met or somebody you argue with. Get round the yard and put a name to most of the shift before the handover.

PLAN CARD 79W

Tuesday, ten past four in the morning. The biggest machine on the Calder system has tripped with no warning. The frequency fell, then steadied below where it started. Dolores Reyes says the steadying matters more than the fall. Nadia Haddad has machines that can cover it, but not in two minutes. Today you read what the frequency is telling you, and decide what to ask for first. Every house on the system is on the other end of it.

BRIEFING 16W

A unit tripped at 04:12 and the frequency has not come back to where it started.

WHAT YOU DECIDE 14W

Establish what the frequency is actually telling the room before anybody acts on it.

1.1 What the falling number tells you

SYSTEM OPERATIONS · A ROOM · PROTOCOL

SCENE 42W

The wall shows the frequency dropping for eight seconds, then flattening out low. Beside it the power is in MW, which means megawatts, millions of watts at once. Aaron Whitlock has the log open, and two people have already named a cause.

GUIDE 58W

Four readings on the left, and what each is evidence of on the right. Pair them by asking one question. Is this one number for the whole system, or one number for a place? Frequency is the same everywhere. Voltage is not. That is what lets one of them say whether something happened, and the other say where.

ASK 9W

Match each reading to what it is evidence of.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

the whole grid runs at one speed, and that speed is what frequency measures

VERDICT 80W

Every generator on the grid turns in step, like a line of dancers. Frequency is the speed they are all turning at, so it is one number for the whole system. Take a generator away and the rest have to carry its share, which slows them all down together. How fast it falls says how big the gap is. Flattening out low is not recovery. It is the machines settling at a new, slower speed with the gap still there.

TAKEAWAY 11W

Frequency belongs to the whole system. Voltage belongs to one place.

1.2 How much went missing

GENERATION · GENERATION HALL · A PERSON · BALLPARK

SCENE 26W

Haddad needs a number before starting anything. The demand has not moved. What has changed is how much the units still running can make between them.

GUIDE 57W

Four numbers, and two of them are about the frequency. Those describe what the shortfall did, not how big it is. Ask of each number whether it is an amount of power. Demand has not moved. What changed is how much the units left running can make. A gap is one of those taken from the other.

ASK 5W

How much generation is missing?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

the grid has to make exactly as much as it is using, every second

VERDICT 73W

A grid has to make exactly what it is using, second by second. There is no store. So when a unit trips, the gap is simply what was needed minus what is left. Here that is three hundred megawatts. The frequency falling is the machines already spinning giving up some of their speed to cover it, which they can do for seconds rather than minutes. That is the whole reason this is urgent.

TAKEAWAY 15W

The gap between what is needed and what is being made is the whole problem.

1.3 What a timestamp is made of

METERING & STANDARDS · A ROOM · CHOICE

SCENE 27W

June Farrow has three records of the same eight seconds, and they disagree about what happened first. The three of them are stamped by two different clocks.

GUIDE 59W

All four options pair two things that would have to be true. Ask of each whether both halves are about this claim. A record saying the alarm came first depends on the alarm happening, and on the clock that stamped it. New wiring and a loud noise are about other things. Three records disagree, and two clocks stamped them.

ASK 19W

A record says the alarm came first. What two things have to be right for that to be true?

BEHIND THE BUTTON 169W

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Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a written-down time is a measurement plus a clock

VERDICT 89W

Every timestamp is two claims glued together. One is that something happened — a relay picked up, a breaker opened. The other is that the clock writing the time was correct at that moment. They fail separately, and that is the whole difficulty. A relay can be perfectly right about what it saw and eighteen months out on when. So when two records disagree about the order, the first question is not which relay is broken. It is which clocks they used, and when those clocks were last checked.

TAKEAWAY 16W

Being sure something happened and being sure when it happened are two separate pieces of work.

END OF THE DAY 20W

Frequency is not a reading from a place; it is the whole system's supply and demand, arriving as a number.

DAY 2 OF 10

What the line can carry

BEFORE THE DAY — TRIAL

Walk the yard before you touch anything

Nobody works on a station they have not walked. Reyes, the shift supervisor, takes you round once in daylight. The control room, the relay house, the transformer bays. You will be sent to all of them this fortnight, and a map means nothing until you have stood there.

PLAN CARD 78W

Wednesday. The replacement power is coming from the wrong side of the system, so it arrives down two lines instead of three. Piotr Novak has a line above its steady limit. A relay will take the decision away from the room in about forty minutes if nobody else does. Thabo Dube is in the yard waiting to be told what to switch. Today you rate the line honestly, and work out where the flow goes if it trips.

BRIEFING 14W

With the unit off, the flow has redistributed and one corridor is running warm.

WHAT YOU DECIDE 11W

Decide which circuit takes the flow once one path is gone.

2.1 Act before the relay does

TRANSMISSION & PROTECTION · A ROOM · TRIGGER

SCENE 35W

The line is carrying eight per cent more than it is rated for, and rising slowly. Novak wants a plan he can hand to the switching operator before the protection makes the choice for them.

GUIDE 72W

One rule, written before the trend arrives. Once you release it the number cannot change. Moving 200 MW off the corridor takes six hours from the moment it is called. At 120 % the protection switches the line off itself. So your line has to sit far enough below 120 that the move is finished first — and not so low that you move power for a corridor that has shown nothing.

ASK 14W

At what loading do you commit the redispatch, when the move takes six hours?

BEHIND THE BUTTON 119W

Why act before the protection. The protection will disconnect the line when its limit is reached, and it will not care what that leaves switched off. Anything you do first is a choice. Anything after is a consequence.

Why two numbers rather than one. Moving generation around takes minutes. Shedding load takes seconds. One threshold set for the slow action fires far too early for the fast one, and set for the fast one leaves no time for the slow one.

What the operator needs. Somebody acting at three in the morning needs a rule, not a judgement. Numbers agreed beforehand are what make the night defensible, and what stop the decision belonging to whoever happens to be awake.

TAKES AS READ

a line has a steady limit and a short-term one, and they are different · moving generation about takes minutes, and shedding load is quick

VERDICT 78W

The short-term limit exists so somebody can decide, not so everybody can wait. The response comes in stages because the actions take different times. Moving generation about takes minutes, so it has to start while there is still room. Shedding load is quick, so it can wait for a higher number. Writing both down before the trend runs stops hindsight becoming policy. A sensible threshold that fires too late for its own action is still a failed rule.

TAKEAWAY 15W

A rule has to fire early enough for the slow action to arrive in time.

2.2 The power that never arrives

LOAD & FORECASTING · A PERSON · BALLPARK

SCENE 34W

Sten Lindgren, who forecasts the demand, wants the loss figure before the morning call. The argument about moving generation about is really an argument about waste. The yard radio is still live beside them.

GUIDE 64W

Four numbers, and two of them belong to other questions. The ohms and the voltage are not in this sum. Ask of each number whether it is what the line carries, or what share of it is wasted. A share is per hundred, so it has to be applied to the amount carried. The waste is heat, and heat is what limits the wire.

ASK 10W

How much of the power is being lost as heat?

BEHIND THE BUTTON 106W

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TAKES AS READ

current pushing through a wire makes heat, and that heat is power that never arrives

VERDICT 75W

Pushing current down a wire warms the wire, and that heat is power that never reaches anybody. About one and a half per cent of what this corridor carries is being spent that way. On eleven hundred and fifty megawatts that is seventeen, which is a small power station keeping the countryside warm. The share is not fixed either. Push more current down the same wire and the waste climbs faster than the current does.

TAKEAWAY 16W

A wire warms up because some of the power is being spent on the wire itself.

2.3 What you are really protecting against

SYSTEM OPERATIONS · A ROOM · CHOICE

SCENE 28W

Reyes wants a decision. The line is inside its short-term limit for another half hour. The question on the table is what happens elsewhere if this one trips.

GUIDE 60W

All four options give a reason to act, and three of them are about this line alone. Ask of each whether it describes what is happening now, or what happens if this line trips. A short-term limit exists so somebody can decide. What makes it dangerous is where the flow goes next. Reyes wants an answer in half an hour.

ASK 11W

What makes the warm line a problem worth acting on now?

BEHIND THE BUTTON 169W

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TAKES AS READ

a network is supposed to survive losing any single line

VERDICT 64W

Running above the steady limit is a decision about time and heat, and the short-term limit exists so somebody can make it. What turns expensive into dangerous is what happens next. This line is already loaded. Lose it, and everything it was carrying lands on its neighbours, which were never sized for the sum. That second trip is protection working properly. Nobody chooses it.

TAKEAWAY 14W

A limit only matters alongside what the flow does when that line is gone.

END OF THE DAY 20W

A rating is a time and a temperature, not a wall, and what matters is where the flow goes next.

DAY 3 OF 10

The last few kilometres

BEFORE THE DAY — FOLLOW

The switching route, walked once by somebody who knows it

Obi, the distribution lead, walks the route in the order the isolations happen, and she will not do it twice. Stay with her. Too far back and you lose which bay she turned into. She stops at each isolation point, and that is where you learn it.

PLAN CARD 70W

Thursday. The big lines are steady, but eleven local feeders are still dead. Chinelo Obi has three crews and fourteen thousand households in the dark. One of her circuits carries a dialysis clinic. Another carries a thousand homes. Today you work out why the far end of a long feeder sits low, and choose which circuit gets the first crew. Every hour of this is somebody sitting in the cold.

BRIEFING 17W

11 feeders are still dead and the crews can only be in 1 place at a time.

WHAT YOU DECIDE 12W

Get customers back on in an order that can be defended afterwards.

3.1 Why the far end sits low

DISTRIBUTION · DISTRIBUTION DEPOT · A ROOM · BALLPARK

SCENE 25W

The substation end of the feeder is at its proper voltage. Customers four kilometres out are complaining. The only meter is back at the substation.

GUIDE 58W

Four numbers, and two of them belong to other questions. The current and the distance are not in this sum. Ask of each number whether it is a drop, or what the feeder should be. A share compares the two. The only meter is back at the substation, and the customers four kilometres out are the ones complaining.

ASK 14W

How big is the drop, as a share of what the feeder should be?

BEHIND THE BUTTON 106W

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TAKES AS READ

current pushing along a wire uses up some of the voltage on the way

VERDICT 70W

Voltage is the push, and pushing current down four kilometres of wire uses some of it up. So the far end always sits lower, and the busier the feeder the lower it goes. Here it drops about a tenth. That is enough for lights to dim and motors to labour, while the meter at the substation reads perfectly normal. The only way to know is to measure at the end.

TAKEAWAY 16W

A voltage measured at one end says nothing about the far end of a loaded line.

3.2 What actually limits a wire

TRANSMISSION & PROTECTION · A PERSON · PROTOCOL

SCENE 29W

Novak has been asked to allow more current down a circuit for the afternoon. He wants each reason for saying no matched to the thing it is really about.

GUIDE 52W

Four observations, and each one is limited by something different. Ask of each whether it is about current, voltage, or where the wire hangs. Heat comes from current and does not care about voltage. Insulation cares about voltage and not about current. One of these four is where heat turns into safety.

ASK 9W

Match each observation to the limit it is about.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

a wire warms up as current passes through it

VERDICT 78W

Two different limits get muddled here. Heat comes from current, and it does not care what voltage the wire is at. The insulation and the gap to anything else care about voltage, and they do not care how much current is flowing. So a circuit can sit at exactly its right voltage and still be too hot. The drop at the far end is a third thing, and it is a customer's problem rather than a safety one.

TAKEAWAY 12W

Current makes the heat. Voltage decides the spacing. They are separate limits.

3.3 The order somebody will read back

SYSTEM OPERATIONS · A ROOM · CHOICE

SCENE 28W

Three crews, eleven dead feeders, and a list in front of Reyes that mixes household counts with a clinic and a pumping station. The phone is already ringing.

GUIDE 51W

Four feeders, and the number of houses is only one of two things that matter. Ask of each what waiting costs, and how long the work takes. A one-hour job frees the crew for the next fault. A four-hour job does not. One of these is already on a standby supply.

ASK 7W

Which feeder should take the first crew?

BEHIND THE BUTTON 169W

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TAKES AS READ

a crew can only work on one fault at a time

VERDICT 67W

Two things decide this, and the number of households is only one. The first is what waiting costs. For a dialysis clinic that is treatments, not inconvenience. The second is how long the work takes. A one-hour job hands the crew back in time for another fault. The clinic circuit is worst on the first count and best on the second, so it goes first either way.

TAKEAWAY 14W

The first crew goes where waiting costs most, not where the count is biggest.

3.4 Can it come off, or not

LOAD & FORECASTING · A ROOM · BELT

SCENE 30W

The order needs a list in ten minutes. Some places can lose power for an hour and be fine. Others have something behind them that cannot simply be switched off.

GUIDE 45W

Two bins. Ask what happens behind the wire while the power is off. If it can wait an hour and come back the same, it can come off. If something behind it is unsafe without power, or cannot be started again on demand, it cannot.

ASK 14W

Send each place to the bin that says whether the order can take it.

BEHIND THE BUTTON 65W

Why big does not mean important. A large factory with its own generator can come off. A small clinic on one supply cannot. Size and money have nothing to do with it.

What makes a place impossible to switch off. Something that fails badly without power, or something that cannot be restarted when you want. A furnace that sets solid. A pump holding water back.

TAKES AS READ

some places can lose power for a while and some cannot

VERDICT 131W

The list is written from what is behind each wire. On one side are the places that lose an hour and are fine afterwards. A cold store stays cold. An office is dark. A works with its own generator carries on. On the other side are places where an hour without power is not an inconvenience. A hospital on one supply. A furnace whose metal sets solid and has to be broken out. A pump holding water out of a town. A machine part way through a run that cannot be stopped. None of that follows from how big a customer is or how much they pay, which is why the list is written in advance and not argued about at four in the morning with somebody shouting down a phone.

TAKEAWAY 12W

What can be switched off depends on what is behind the wire.

3.5 How much went missing

GENERATION · GENERATION HALL · A ROOM · CALLBACK · BALLPARK

SCENE 26W

Haddad needs a number before starting anything. The demand has not moved. What has changed is how much the units still running can make between them.

GUIDE 57W

Four numbers, and two of them are about the frequency. Those describe what the shortfall did, not how big it is. Ask of each number whether it is an amount of power. Demand has not moved. What changed is how much the units left running can make. A gap is one of those taken from the other.

ASK 5W

How much generation is missing?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

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TAKES AS READ

the grid has to make exactly as much as it is using, every second

VERDICT 73W

A grid has to make exactly what it is using, second by second. There is no store. So when a unit trips, the gap is simply what was needed minus what is left. Here that is three hundred megawatts. The frequency falling is the machines already spinning giving up some of their speed to cover it, which they can do for seconds rather than minutes. That is the whole reason this is urgent.

TAKEAWAY 15W

The gap between what is needed and what is being made is the whole problem.

END OF THE DAY 16W

Restoration is a series of choices about people, made with instruments that only see the substation.

Is that number the number

PLAN CARD 76W

Friday. Tuesday night has to be written down, and Dolores Reyes wants it filed today. June Farrow will not sign it. Two of its figures come from instruments last checked in the spring. Nadia Haddad needs a machine back on the bars before the evening peak, and there is a right order for that. Today you work out what the system was really delivering, and settle an argument between two instruments. Farrow is right this time.

BRIEFING 17W

The report is due and two of the figures in it come from instruments nobody has checked.

WHAT YOU DECIDE 11W

Establish which readings the incident account is allowed to rest on.

4.1 Power for four hours

METERING & STANDARDS · A ROOM · BALLPARK

SCENE 31W

Farrow has what the circuit was delivering on the night of the event, and how long it ran for. The report claims an energy figure nobody has been able to reproduce.

GUIDE 54W

Four numbers, and two of them are the voltage and the current. Those were used to work out the power already. Ask which two the question needs. Power says how fast energy arrives. Energy says how much arrived, and it needs a length of time. A report already claims a figure nobody can reproduce.

ASK 9W

How much energy did the circuit deliver that night?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

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TAKES AS READ

power says how fast energy is being delivered, and energy is what actually gets used

VERDICT 69W

Power and energy get muddled constantly, and they are not the same thing. Power says how fast. Five and a half megawatts is a rate, like a tap running. Energy is what came out of the tap, and it depends on how long the tap was open. Four hours at 5.6 megawatts is 22.4 megawatt hours. It is the energy that gets billed, and the energy that runs out.

TAKEAWAY 18W

Power is a rate. Energy is what the rate delivers over time, and the bill is in energy.

4.2 Three things that have to agree

GENERATION · GENERATION HALL · A PERSON · SEQUENCE

SCENE 30W

Haddad has a machine at rest and a grid running normally. She wants the order written down before anybody is standing at the panel with a hand on the switch.

GUIDE 61W

All three of these happen at the panel, so nothing is missing from the list. Ask of each what difference it removes. Then ask what that difference would do if the switch closed while it was still there. A speed difference and a voltage difference break things in different ways. One of these three only works once the others are right.

ASK 12W

Put the steps of joining a machine to the grid in order.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it. Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

a generator joined to a live grid is forced to turn at the grid's speed

VERDICT 74W

Closing that switch ties the machine to the whole grid, and the grid does not adjust. So every difference has to go first. Speed first, because a machine turning at the wrong speed gets slammed into step, hard enough to damage it. Voltage next, or current pours between them for no useful reason. Then wait for the two waves to line up. Close them when all three match, and the machine simply joins in.

TAKEAWAY 16W

Every mismatch gets removed before the switch closes, and there is only one order that works.

4.3 Two instruments, two numbers

TRANSMISSION & PROTECTION · A ROOM · CHOICE

SCENE 29W

A scope on the busbar shows the voltage peaking at 325 volts. The panel meter beside it reads 230. Both were checked last month and neither has been touched.

GUIDE 59W

Four relations, and both instruments are working. So ask what each number is a summary of. The voltage is changing all the time, so there is no single value. One instrument reports the highest it reaches. The other reports the steady voltage that would heat something the same. Two of these options have the conversion the wrong way round.

ASK 9W

Which relation should hold if both instruments are right?

BEHIND THE BUTTON 169W

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TAKES AS READ

an alternating voltage rises and falls, over and over, rather than sitting still

VERDICT 81W

An alternating voltage is never one number. It rises to a peak, falls back through zero, and does the same the other way, fifty times a second. The scope shows the highest it reaches. The meter shows the steady voltage that would do the same work — heat the same kettle in the same time. For this shape of wave the peak sits about one and a half times higher, and 325 against 230 is exactly that. Two instruments, one voltage.

TAKEAWAY 18W

An alternating voltage has no single value, so the number quoted has to say which one it is.

4.4 What the falling number tells you

SYSTEM OPERATIONS · A ROOM · CALLBACK · PROTOCOL

SCENE 42W

The wall shows the frequency dropping for eight seconds, then flattening out low. Beside it the power is in MW, which means megawatts, millions of watts at once. Aaron Whitlock has the log open, and two people have already named a cause.

GUIDE 58W

Four readings on the left, and what each is evidence of on the right. Pair them by asking one question. Is this one number for the whole system, or one number for a place? Frequency is the same everywhere. Voltage is not. That is what lets one of them say whether something happened, and the other say where.

ASK 9W

Match each reading to what it is evidence of.

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Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

the whole grid runs at one speed, and that speed is what frequency measures

VERDICT 80W

Every generator on the grid turns in step, like a line of dancers. Frequency is the speed they are all turning at, so it is one number for the whole system. Take a generator away and the rest have to carry its share, which slows them all down together. How fast it falls says how big the gap is. Flattening out low is not recovery. It is the machines settling at a new, slower speed with the gap still there.

TAKEAWAY 11W

Frequency belongs to the whole system. Voltage belongs to one place.

END OF THE DAY 17W

A reading is a claim about one quantity, and two instruments can disagree while both are right.

DAY 5 OF 10

Enough in the tank

BEFORE THE DAY — HUNT

Six earths, and the log says five

Before the line goes live again every temporary earth has to come off. The log says five went on. Lindgren, the load forecasting analyst, counted six. Nothing gets switched in until all of them are found, because an earth left on a live line is a huge fault.

PLAN CARD 74W

Monday. The forecast for tomorrow's evening peak has moved four degrees colder. Sten Lindgren says the middle has hardly changed, but the width has, and the width is what the reserve must cover. Rafael Alvarez has weekend storage figures that are worse than the brochure. Today you work out what the battery will really give back, and decide what is worth finding out before the peak. Hold too little back and Thursday happens again.

BRIEFING 12W

Tomorrow's peak is colder than the forecast the reserve was set against.

WHAT YOU DECIDE 12W

Decide what the system is holding back, and whether it is enough.

5.1 What comes out against what went in

LOAD & FORECASTING · A ROOM · BALLPARK

SCENE 27W

The battery charged overnight on Saturday and gave it back into Sunday's peak. Alvarez has both totals and the brochure figure, and the three do not agree.

GUIDE 62W

Four numbers, and two of them are the power rating and how long it can run. Those say how fast and for how long, not how much came back. Ask of each whether it is an amount of energy. A round trip compares what went in with what came out. Three numbers on the desk disagree, and one is from a brochure.

ASK 10W

What share of the energy put in comes back out?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

putting energy into a store and taking it out again both waste some of it

VERDICT 79W

Nothing stores energy for free. Some is lost going in, some coming out, and some keeping the thing cool while it sits there. So the number that matters is simply what came out divided by what went in. Here that is 187 out of 220, which is about eighty-five per cent. Fifteen per cent of everything that went in never came back. The power rating on the label is a different claim. It says how fast, not how much.

TAKEAWAY 17W

What a store is worth is the share that comes back, not the size on the label.

5.2 Why the wires are not at wall voltage

GENERATION · GENERATION HALL · A PERSON · CHOICE

SCENE 22W

A visitor asks Haddad why the machine makes power at twenty thousand volts and the line outside runs at four hundred thousand.

GUIDE 59W

All four options pair a change in current with a change in waste. Two of them get the current right and disagree about the waste. Ask what the waste depends on, and how strongly. Current falls in step with the voltage rising. Waste follows the current squared. Work out both halves, because three of these get one half right.

ASK 13W

For the same power, what happens when the voltage is raised twenty times?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

for the same power, a higher voltage means less current

VERDICT 75W

Power is the push multiplied by the flow. So for the same power, a bigger push needs less flow. Raise the voltage twenty times and the current falls to a twentieth. The waste in the wires depends on the current twice over, so a twentieth of the current wastes four hundred times less. That is why long lines run at enormous voltages. The insulation is harder and more expensive, and the saving is far bigger.

TAKEAWAY 18W

The same power can be delivered in more than one way, and the ways waste very different amounts.

5.3 What is worth knowing before the peak

SYSTEM OPERATIONS · A ROOM · VALUE

SCENE 28W

Reyes has a small budget and four things she could find out before the evening peak. Only some of them would change what she does about the reserve.

GUIDE 49W

Five credits, four things you could find out, and a decision at six o'clock. Open each card and ask what its result would change. A measurement that confirms something already decided is not worth a credit. Buy the ones that could still change what Reyes does about the reserve.

ASK 8W

Which measurements are worth buying before six o'clock?

BEHIND THE BUTTON 122W

What makes evidence worth buying. Two possible results that would lead to different actions. If both answers leave you doing the same thing, the measurement is interesting and not useful, and tonight that difference is a credit you cannot get back.

Why the clock is part of the question. A result that arrives after six is a result about a decision already made. So a rough answer in time can be worth more than a careful one that is late.

What the reserve is. Spare generation held back so it can be used if something fails. Deciding how much to hold is a bet on what might go wrong, which is why evidence about what might go wrong is what changes it.

TAKES AS READ

reserve is held against what nobody knows, rather than against the forecast itself

VERDICT 74W

Reserve is held against two things. How wrong the demand forecast could be, and whether the things being counted on can actually deliver. Measuring how this winter's cold has been driving demand narrows the first. Testing the store at the rate it would really be called on settles the second. Repeating a generator test that is already valid buys reassurance. A survey that lands in six weeks cannot change tonight, however good it is.

TAKEAWAY 14W

A measurement is worth buying when its answer could still change what you do.

5.4 What a timestamp is made of

METERING & STANDARDS · A ROOM · CALLBACK · CHOICE

SCENE 27W

June Farrow has three records of the same eight seconds, and they disagree about what happened first. The three of them are stamped by two different clocks.

GUIDE 59W

All four options pair two things that would have to be true. Ask of each whether both halves are about this claim. A record saying the alarm came first depends on the alarm happening, and on the clock that stamped it. New wiring and a loud noise are about other things. Three records disagree, and two clocks stamped them.

ASK 19W

A record says the alarm came first. What two things have to be right for that to be true?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a written-down time is a measurement plus a clock

VERDICT 89W

Every timestamp is two claims glued together. One is that something happened — a relay picked up, a breaker opened. The other is that the clock writing the time was correct at that moment. They fail separately, and that is the whole difficulty. A relay can be perfectly right about what it saw and eighteen months out on when. So when two records disagree about the order, the first question is not which relay is broken. It is which clocks they used, and when those clocks were last checked.

TAKEAWAY 16W

Being sure something happened and being sure when it happened are two separate pieces of work.

END OF THE DAY 19W

A reserve is bought with something, and the argument is always about how much uncertainty it has to cover.

DAY 6 OF 10

80 milliseconds

PLAN CARD 71W

Wednesday. A cable failed inside the substation at twenty to two. The protection cleared it in under a tenth of a second, so nobody in the control room saw the fault — only the aftermath. Piotr Novak wants the fault current worked out before he signs the switchgear back into service. Ewa Kowalczyk has a crew at the gate. Today you size the fault, and decide who may go near it.

BRIEFING 17W

A cable failed at the substation and the protection cleared it before the alarm reached the room.

WHAT YOU DECIDE 11W

Establish what a fault does before anybody can respond to it.

6.1 What a short circuit is limited by

TRANSMISSION & PROTECTION · A ROOM · BALLPARK

SCENE 34W

The cable failed and shorted to earth. The push behind it is 6,350 volts, and the whole path back to the source resists 0.42 ohms. Novak wants the current the switchgear had to break.

GUIDE 57W

Four numbers, and two of them belong to the healthy circuit. The normal load is set by equipment using power, and a fault goes around that equipment. Ask of each number whether it describes the load or the path back to the source. The clearing time says how long the current lasted, not how big it was.

ASK 7W

How much current flowed during the fault?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a short circuit is a path with almost nothing in the way

VERDICT 64W

Normally the current is set by whatever is drawing power at the far end. A short circuit goes round all of that. So the only things left holding it back are the wire and the machinery behind it. Push divided by resistance is fifteen thousand amps, about fifty times the normal load. That is why switchgear has its own rating for breaking a fault.

TAKEAWAY 16W

A fault current is set by the path back to the source, and by nothing else.

6.2 A reading with no connection

METERING & STANDARDS · A PERSON · CONTROL

SCENE 36W

June Farrow has a safe low-voltage test rig beside the yard diagram. It shows the current going in, the signal coming out, and a core she can open. She wants a cause rather than a definition.

GUIDE 69W

The rig is a current transformer. A wire carries current, and a ring of iron around it holds a coil that reports a copy of it. The number you watch is what that coil reports. Three controls can be changed. A change counts only if the number moves more than it wanders on its own. Change one, run the test, undo it, and name the control the reading follows.

ASK 15W

Which change makes the signal disappear, and does it come back when you undo it?

BEHIND THE BUTTON 160W

How the rig works, and why nothing touches the wire. Current in the wire makes a magnetic field. The iron ring carries that field around the coil. When the field changes, it pushes a current through the coil. Nothing is connected to the wire itself, which is what makes the meter safe to use on a live line.

Why a steady current reads nothing. The coil only answers to a field that is changing. Hold the current still and the field stops changing, so the coil reports nothing while the wire is still carrying full load. The meter looks broken and is working exactly as designed.

Why the iron ring is one of the controls. Open the ring and the field no longer reaches the coil. The reading falls away even though the current is still changing. So two different changes can kill the reading for two different reasons, and putting each one back is how you tell them apart.

TAKES AS READ

a changing magnetic field can make a voltage appear in a nearby coil

VERDICT 59W

There is no wire between the two sides of this device. What crosses is a magnetic field, made by the current going in. A field on its own does nothing to the coil. It has to be changing. Alternating current changes direction fifty times a second, so a signal appears. Steady current makes a steady field and nothing else.

TAKEAWAY 17W

What makes a signal appear here is change. A steady current makes a field and no signal.

6.3 Isolated is not yet safe

DISTRIBUTION · DISTRIBUTION DEPOT · A ROOM · CHOICE

SCENE 32W

The cable is cut off at both ends and a tester reads dead at the work point. Kowalczyk stops the crew, because one step is still missing. There is live equipment nearby.

GUIDE 59W

All four options are things a crew could do next, and three of them sound careful. Ask of each what it would protect against. Something you already know about, or something that arrives later? Cutting it off removes the known sources. A dead test proves one moment. Neither of them holds the cable down if a switch moves nearby.

ASK 9W

What has to happen after isolating and testing dead?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a conductor can be brought back to life from more than one direction

VERDICT 58W

Isolating and testing answer different questions, and neither answers the one that matters. Opening both ends removes the sources anybody knows about. The tester says it is dead now. Neither promises it stays dead if somebody switches something. Earthing gives any surprise somewhere easy to go, instead of through a person. It turns a fact into a condition.

TAKEAWAY 12W

Isolating and testing say what is true now. Earthing keeps it true.

6.4 The power that never arrives

LOAD & FORECASTING · A ROOM · CALLBACK · BALLPARK

SCENE 34W

Sten Lindgren, who forecasts the demand, wants the loss figure before the morning call. The argument about moving generation about is really an argument about waste. The yard radio is still live beside them.

GUIDE 64W

Four numbers, and two of them belong to other questions. The ohms and the voltage are not in this sum. Ask of each number whether it is what the line carries, or what share of it is wasted. A share is per hundred, so it has to be applied to the amount carried. The waste is heat, and heat is what limits the wire.

ASK 10W

How much of the power is being lost as heat?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

current pushing through a wire makes heat, and that heat is power that never arrives

VERDICT 75W

Pushing current down a wire warms the wire, and that heat is power that never reaches anybody. About one and a half per cent of what this corridor carries is being spent that way. On eleven hundred and fifty megawatts that is seventeen, which is a small power station keeping the countryside warm. The share is not fixed either. Push more current down the same wire and the waste climbs faster than the current does.

TAKEAWAY 16W

A wire warms up because some of the power is being spent on the wire itself.

END OF THE DAY 19W

A fault is limited only by impedance, which is why the numbers are enormous and the times are tiny.

DAY 7 OF 10

Merit order

BEFORE THE DAY — CANVASS

Whose lights are actually back

The system says eleven feeders are restored. The complaint line says otherwise. A feeder that reads live at the station and dead at the street has a fault in between. Ask along the streets until enough answers have come back.

PLAN CARD 73W

Thursday. Tomorrow has to be bought tonight. Sten Lindgren has a peak that moves with the temperature and a width he will not narrow. Mina Sarraf has a ridge that will produce almost nothing for money and cannot promise any of it. Nadia Haddad has expensive machines that do exactly as they are told. Today you forecast the peak, and decide what cheapest-first really means when the cheapest plant is the least certain.

BRIEFING 18W

The cheapest generation is on the ridge and it will not say what it is doing after four.

WHAT YOU DECIDE 15W

Decide what runs tomorrow, and what is held back for the day it is needed.

7.1 The peak, and what the cold adds

LOAD & FORECASTING · A ROOM · BALLPARK

SCENE 31W

A normal Friday this week of the year peaks at 6,400 megawatts. Lindgren's own figure for this winter is 120 megawatts more for every degree below zero. Tomorrow is minus four.

GUIDE 60W

Four numbers, and one of them is the reserve, which is about how wrong the forecast might be. Ask of each whether it is a normal peak, an amount per degree, or a number of degrees. An amount per degree has to be multiplied by the degrees. What comes out is the middle of a forecast, not the worst case.

ASK 6W

What should tomorrow's evening peak be?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

demand goes up as the temperature goes down, roughly in step

VERDICT 77W

Demand looks much the same most days, because people do much the same things at much the same times. What moves it is weather. The useful way to hold that is a figure for what one degree is worth on this system, measured from this winter rather than taken from a book. Here it is a hundred and twenty megawatts a degree, and tomorrow is four degrees below. So the peak lands near six thousand nine hundred.

TAKEAWAY 17W

A forecast is a normal shape plus what the weather adds, and the second part gets measured.

7.2 Cheapest first, and the exceptions

SYSTEM OPERATIONS · A PERSON · CASEBOOK

SCENE 26W

Four plants and an order to sign. Reyes wants each one matched to the real reason it sits where it does, rather than to its price.

GUIDE 51W

Four plants and four reasons for their place in the queue. Only one of those reasons is price. Pair them by asking what else matters: whether the output can be promised, whether the machine can move fast, or where it sits. A machine at full output has nothing spare to offer.

ASK 10W

Match each plant to why it sits where it does.

BEHIND THE BUTTON 153W

Why a clue needs an explanation and not a name. Saying what a clue is called is not saying what produced it. The board makes you commit to a cause, and a cause has to account for that clue and not for the one next to it. That is the difference between recognising something and explaining it.

How to use the one-each rule. Every explanation goes with exactly one clue. So the pairs you are sure about are worth more than themselves — each one you settle leaves fewer explanations for the clues you are unsure of. Two you trust can decide the other two.

Why one wrong line is impossible. If three lines are right, the last explanation has nowhere else to go, so it is right too. Being wrong always means two are wrong at once. When a join feels forced, look again at one you made early and stopped questioning.

TAKES AS READ

generators are normally run cheapest first

VERDICT 70W

Running the cheapest first is what keeps electricity affordable. Three things bend that order. A plant that costs almost nothing to run goes first, and if nobody can promise its output, something reliable stays behind it. A plant that can change output fast is worth holding at half, because the other half is what gets called on. And a plant in the right place can be needed whatever it costs.

TAKEAWAY 25W

Cheapest first is the rule, and three things bend it: where a plant is, how fast it moves, and whether it can be relied on.

7.3 When the cheapest power cannot get out

GENERATION · GENERATION HALL · A ROOM · SCIENCETANK

SCENE 24W

Sarraf can make 340 megawatts on the ridge tonight. The line out of the valley carries about 190. There is no second way out.

GUIDE 98W

The ridge can make 340 megawatts tonight. The line out is rated at about 190. Nights like this happen about 40 times a winter. A store on the ridge would be 50 megawatts for four hours, and it gives back 85 of every 100 units put in. The second circuit has been on the list for two years with no date, and takes about three years once it starts. Curtailment payments come out of the same budget every night like this one. Running a line above its rating heats the wire, and heat is what makes it sag.

ASK 12W

Decide what to do with more wind than the line can carry.

BEHIND THE BUTTON 147W

Why every number is graded. Funding is not a vote for one idea. Where you put the small numbers says what you think is worth a hedge and what is not worth doing at all. Backing the right proposal and quietly funding a bad one is still a bad answer.

What the three numbers are for. Thirty-five is what makes a lead a lead. Below it you have hedged instead of chosen. Fifteen is the most that can sit on work the evidence does not back before it counts as a second opinion. Twenty keeps strong work funded well enough to finish, because half-funding it spends the money and buys nothing.

Why you cannot hold points back. Unspent points are not caution. They are a decision not to decide, made with somebody else's money and somebody else's deadline. The floor is what makes the panel say something.

TAKES AS READ

power has to have a path to wherever it is used

VERDICT 80W

A line at its limit is a wall, not a preference, so some of tonight's wind is not going anywhere. The only question is what the system buys instead. A second line fixes it for good and takes years. A store on the ridge moves the surplus in time rather than in space, which is the right shape for a problem that happens forty nights a winter. Paying the fleet to stop is what happens meanwhile, and it buys nothing.

TAKEAWAY 13W

Electricity that cannot reach anybody is not electricity, whatever it cost to make.

7.4 **Act before the relay does**

TRANSMISSION & PROTECTION · A ROOM · CALLBACK · TRIGGER

SCENE 35W

The line is carrying eight per cent more than it is rated for, and rising slowly. Novak wants a plan he can hand to the switching operator before the protection makes the choice for them.

GUIDE 72W

One rule, written before the trend arrives. Once you release it the number cannot change. Moving 200 MW off the corridor takes six hours from the moment it is called. At 120 % the protection switches the line off itself. So your line has to sit far enough below 120 that the move is finished first — and not so low that you move power for a corridor that has shown nothing.

ASK 14W

At what loading do you commit the redispatch, when the move takes six hours?

BEHIND THE BUTTON 119W

Why act before the protection. The protection will disconnect the line when its limit is reached, and it will not care what that leaves switched off. Anything you do first is a choice. Anything after is a consequence.

Why two numbers rather than one. Moving generation around takes minutes. Shedding load takes seconds. One threshold set for the slow action fires far too early for the fast one, and set for the fast one leaves no time for the slow one.

What the operator needs. Somebody acting at three in the morning needs a rule, not a judgement. Numbers agreed beforehand are what make the night defensible, and what stop the decision belonging to whoever happens to be awake.

TAKES AS READ

a line has a steady limit and a short-term one, and they are different · moving generation about takes minutes, and shedding load is quick

VERDICT 78W

The short-term limit exists so somebody can decide, not so everybody can wait. The response comes in stages because the actions take different times. Moving generation about takes minutes, so it has to start while there is still room. Shedding load is quick, so it can wait for a higher number. Writing both down before the trend runs stops hindsight becoming policy. A sensible threshold that fires too late for its own action is still a failed rule.

TAKEAWAY 15W

A rule has to fire early enough for the slow action to arrive in time.

END OF THE DAY 18W

Dispatch orders plant by cost, and the value of certainty is what stops that being the whole story.

DAY 8 OF 10

Cranking path

BEFORE THE DAY — EVADE

The contractor who wants a decision in the yard

A contractor has been here since six wanting an answer about the metering hut. Reyes, the shift supervisor, has said nothing is agreed on the ground without paperwork. Keep out of his way and get your own round done.

PLAN CARD 75W

Saturday evening. The island has gone, the valley is dark, and every machine in it needs electricity before it can start. The pumps, the fans, the controls, the switchgear itself. Nadia Haddad has one small water-driven machine that can start on its own. Thabo Dube has a yard full of open switches. Today you work out why that small machine matters more than the big one, and write the restart in the order physics allows.

BRIEFING 17W

The valley collapsed at 19:04 and nothing in it can start without power it has not got.

WHAT YOU DECIDE 15W

Build a live system out of a dead one, in the only order that works.

8.1 What the dark hours cost

GENERATION · GENERATION HALL · A ROOM · BALLPARK

SCENE 30W

The valley was drawing about 300 megawatts when it went dark. Haddad expects two and a half hours to bring it back, and the report wants energy rather than power.

GUIDE 58W

Four numbers, and two of them belong to other questions. The household count and the frequency are not in this sum. Ask of each whether it is a power or a length of time. Power says how fast. Energy says how much, and it grows with every hour. Both get called demand, which is where this goes wrong.

ASK 8W

How much energy did the valley never get?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

power kept up over a stretch of time delivers energy

VERDICT 71W

Power says how fast electricity is arriving. Energy says how much arrived. Both get called demand, which is why they get muddled. Multiply the power by the hours it was missing and you get energy: three hundred megawatts for two and a half hours is seven hundred and fifty megawatt hours. That is the figure every claim and every inquiry runs on, because it grows with the length of the outage.

TAKEAWAY 14W

Power is the rate. Energy is the amount, and everything is settled in energy.

8.2 The order physics allows

SYSTEM OPERATIONS · A ROOM · CHAIN

SCENE 32W

One hydro machine that can start with no outside power. A dead network. And a room that wants the big station back first. Nothing gets switched until the order is written down.

GUIDE 57W

Build the path in the order the power really travels, starting from the machine that can start with no outside power. Then name the required link that decides it. The biggest station is not the answer, and neither is the largest number on the board. A path is limited by the link with the least to give.

ASK 13W

How does starting power reach the big station, and which link decides it?

BEHIND THE BUTTON 141W

Why most machines cannot start themselves. A generator needs power before it can make power: for pumps, for controls, for the magnets that make the field. In a blackout there is none. A hydro machine with a battery and water behind it does not need any, so it is where the network has to begin.

Why the order is not a preference. Switching a path on means feeding the lines and the machines along it in an order the equipment can survive. Ask for too much at once and the small starting machine stalls, and everything is dark again with less left to try.

Why the room wants the wrong thing. The big station is what everybody is waiting for. That is a reason to write the order down before anybody touches a switch, not a reason to put it first.

TAKES AS READ

the big station needs electricity for its own pumps and fans before it can start · the hydro machine can start with no supply at all

VERDICT 73W

A dead network cannot start from its biggest generator, because that machine needs power before it can make any. Pumps, fans, controls. The hydro machine is different: water pushes the turbine round with no electricity involved at all. So it goes first, and its output travels down a small feeder to wake the big station's own equipment. That small feeder is the link everything waits on. It only has to carry the auxiliaries.

TAKEAWAY 12W

The biggest machine is useless until something smaller can wake it up.

8.3 Why it cannot go faster

DISTRIBUTION · DISTRIBUTION DEPOT · A ROOM · CASEBOOK

SCENE 29W

Obi has crews asking why the restart is moving a few streets at a time. The substation is live and the whole feeder could be closed in one movement.

GUIDE 57W

Four steps of the restart, and four different limits. Pair them by asking what is small at that moment. The live system? The load on the line? The agreement between two systems? One of these is the opposite of what most people expect. A long line with nothing on it pushes the far end up, not down.

ASK 10W

Match each step of the restart to what limits it.

BEHIND THE BUTTON 153W

Why a clue needs an explanation and not a name. Saying what a clue is called is not saying what produced it. The board makes you commit to a cause, and a cause has to account for that clue and not for the one next to it. That is the difference between recognising something and explaining it.

How to use the one-each rule. Every explanation goes with exactly one clue. So the pairs you are sure about are worth more than themselves — each one you settle leaves fewer explanations for the clues you are unsure of. Two you trust can decide the other two.

Why one wrong line is impossible. If three lines are right, the last explanation has nowhere else to go, so it is right too. Being wrong always means two are wrong at once. When a join feels forced, look again at one you made early and stopped questioning.

TAKES AS READ

a small live system is upset far more by a given block of load than a big one

VERDICT 79W

What limits the restart changes as it grows. Very early, the live piece is tiny, so any block of load is enormous next to it and the frequency dives. Long lines with almost nothing on them push the voltage up rather than down, which is the opposite of most people's instinct. Streets that have been off in winter come back higher than normal, because every heater is calling at once. And joining two live systems is a matching problem.

TAKEAWAY 15W

Every step is limited by something different, and rushing the wrong one collapses it again.

8.4 Fifty, through the evening

SYSTEM OPERATIONS · A PERSON · HOLD

SCENE 27W

The evening rise is starting and the number on the wall is drifting down with it. The regulating set is the one control left on the desk.

GUIDE 50W

Hold the frequency inside the band on the display. The band narrows as the evening goes on, because there is less spare generation at the peak. Each block of load that comes on keeps pulling until generation answers it. Set the control where it matches, not where it catches up.

ASK 9W

Hold the system at fifty through the evening rise.

BEHIND THE BUTTON 63W

Why the number moves at all. Frequency is the balance between what is being made and what is being used. More being used than made, and the machines slow down.

Why a block of load is not one push. It keeps drawing power for as long as it is switched on. So the frequency keeps falling until generation is raised to match it.

TAKES AS READ

frequency falls when more power is used than is being made

VERDICT 126W

Frequency is a balance. When more power is being used than made, the machines slow down and the number falls. Every block of load in this evening keeps drawing for as long as it is on, so the number keeps falling until generation is raised to match. That is why the control is moved to a new setting and left there. Nudging it and letting go is chasing a needle rather than fixing a mismatch. The band narrows later because there is less spare generation spinning at the peak, so the same drop is closer to the point where protection starts switching customers off by itself. And that is the thing nobody wants: the system does the shedding for you, and it does not choose who.

TAKEAWAY 13W

A mismatch you do not answer does not go away on its own.

8.5 What actually limits a wire

TRANSMISSION & PROTECTION · A ROOM · CALLBACK · PROTOCOL

SCENE 29W

Novak has been asked to allow more current down a circuit for the afternoon. He wants each reason for saying no matched to the thing it is really about.

GUIDE 52W

Four observations, and each one is limited by something different. Ask of each whether it is about current, voltage, or where the wire hangs. Heat comes from current and does not care about voltage. Insulation cares about voltage and not about current. One of these four is where heat turns into safety.

ASK 9W

Match each observation to the limit it is about.

BEHIND THE BUTTON 161W

Why you match instead of choosing. A response that is right for one situation is often nearly right for the next one. Picking from a list lets you take the nearly-right answer and never notice. Matching makes you say which situation it fits best, and that comparison is the whole point.

How to use the one-each rule. Every response goes somewhere, and nowhere twice. So the situations you are sure about are worth more than themselves: each one you settle leaves fewer responses for the ones you are not sure about. Two lines you trust can decide the other two for you.

Why one wrong line is impossible. If three lines are right, the fourth response has nowhere else to go, so it is right too. Being wrong always means at least two are wrong at once. If a join feels forced, the mistake is probably not in that join — it is in one you already made and stopped questioning.

TAKES AS READ

a wire warms up as current passes through it

VERDICT 78W

Two different limits get muddled here. Heat comes from current, and it does not care what voltage the wire is at. The insulation and the gap to anything else care about voltage, and they do not care how much current is flowing. So a circuit can sit at exactly its right voltage and still be too hot. The drop at the far end is a third thing, and it is a customer's problem rather than a safety one.

TAKEAWAY 12W

Current makes the heat. Voltage decides the spacing. They are separate limits.

END OF THE DAY 18W

A dead system is rebuilt from one machine that needs nothing, outward along a path chosen in advance.

DAY 9 OF 10

Three things at once

PLAN CARD 77W

Thursday, twenty to five, and three things arrive together. A cable fault has taken a feeder with three thousand three hundred households on it. Ravi Lindgren's forecast is running two hundred megawatts under, and climbing. And the gas machine that has carried the peak all week must come off at six. Dolores Reyes has one crew and about eighty minutes. Today you decide what gets attention first. None of these is new; all three at once is.

BRIEFING 17W

A fault, a forecast miss and a plant coming off at the same hour, and one room.

WHAT YOU DECIDE 11W

Rank three simultaneous problems, and hold the reasoning while they interfere.

9.1 Which of the three cannot wait

SYSTEM OPERATIONS · A ROOM · DELEGATE

SCENE 33W

Reyes has three live problems on the board and can only keep one herself. Whitlock is ready to take a handover, and she will not accept an instruction that amounts to watch it.

GUIDE 56W

Three problems, and you keep one. For each of the other two, name who takes it, the first real thing they will do, and the reading or event that brings it back to you. Reyes will not accept anything that amounts to watch it. Then take the watch on the one that needs your own judgement.

ASK 11W

What do you keep, and what exactly do you hand over?

BEHIND THE BUTTON 126W

Why watch it fails. Somebody told to observe will come back for orders at the moment you are busiest. A first action is what lets them hold the problem without you, and naming it is the difference between handing something over and putting it off.

What a return condition is. Not a promise to check in. It is the number or event at which the problem stops being theirs, agreed now, so nobody has to decide in the moment whether this is bad enough to interrupt you.

How to choose what to keep. Ask which is getting worse fastest, which is already contained, and what happens if nobody acts. The one that needs a judgement rather than a procedure is the one you cannot hand over.

TAKES AS READ

a bad problem that is not getting worse is different from a small one that is · getting reserve in place takes time before the peak

VERDICT 71W

The cable fault is loud and stable. Those customers are already off, and a crew is on the way. The machine running out of fuel has a known time and a known size, so somebody else can line up its replacement. The forecast error is different. It is still growing, and getting reserve in place takes hours. Every update that goes by without a decision removes choices nobody can get back.

TAKEAWAY 18W

Urgency is about how fast something is getting worse and whether somebody else can own the next move.

9.2 The same line, busier

TRANSMISSION & PROTECTION · A PERSON · BALLPARK

SCENE 31W

With the feeder out, the flow has shifted and the line is carrying more than it was. Novak wants the heat figure before deciding whether the evening needs a switching change.

GUIDE 60W

Four numbers, and two of them describe an hour ago. Ask of each whether it is about the line now. A share of waste has to go with the amount carried at the same time. Both have gone up. That is why a small rise in what the line carries is a bigger rise in the heat than it looks.

ASK 9W

How much is the line wasting as heat now?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

the busier a wire is, the larger the share of the power it turns into heat

VERDICT 76W

An hour ago this line carried eleven hundred and fifty megawatts and wasted about seventeen. Now it carries twelve hundred and eighty and wastes about twenty. The flow went up about a ninth. The waste went up about a fifth. That is because the heat depends on the current twice over, so it always outruns the flow. And the limit on the line is a heat limit, so the margin disappears faster than the loading suggests.

TAKEAWAY 17W

When something counts twice over, a small rise in it is a big rise in the result.

9.3 An order that survives being carried out

LOAD & FORECASTING · A ROOM · SEQUENCE

SCENE 31W

Lindgren has three moves for the evening and no view on the order. One of them changes what the other two are worth, and one cannot be undone inside the shift.

GUIDE 60W

All three moves could be made right now, so this is not a clock. Ask of each two things. What does it improve about the numbers you are working from? And how easily can it be undone? One of these sizes every other figure in the shift. One of them cannot be taken back once the shift has moved on.

ASK 7W

Put the evening's four moves in order.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it. Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

some actions change what the actions after them are worth

VERDICT 67W

Fix the forecast first. Every other number is sized against it. Get it wrong and you do the right thing in the wrong amount. Start loading the replacement plant next. It needs real minutes, and it has to be under way before the fuel runs down. Reconfiguring the network goes last. Nobody can quietly undo it, so it waits until the numbers behind it have stopped moving.

TAKEAWAY 12W

Do what improves your information first, and what you cannot undo last.

9.4 Why the far end sits low

DISTRIBUTION · DISTRIBUTION DEPOT · A ROOM · CALLBACK · BALLPARK

SCENE 25W

The substation end of the feeder is at its proper voltage. Customers four kilometres out are complaining. The only meter is back at the substation.

GUIDE 58W

Four numbers, and two of them belong to other questions. The current and the distance are not in this sum. Ask of each number whether it is a drop, or what the feeder should be. A share compares the two. The only meter is back at the substation, and the customers four kilometres out are the ones complaining.

ASK 14W

How big is the drop, as a share of what the feeder should be?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

current pushing along a wire uses up some of the voltage on the way

VERDICT 70W

Voltage is the push, and pushing current down four kilometres of wire uses some of it up. So the far end always sits lower, and the busier the feeder the lower it goes. Here it drops about a tenth. That is enough for lights to dim and motors to labour, while the meter at the substation reads perfectly normal. The only way to know is to measure at the end.

TAKEAWAY 16W

A voltage measured at one end says nothing about the far end of a loaded line.

END OF THE DAY 22W

Ranking is not about which problem is worst but about which one gets worse fastest and which one is cheapest to remove.

What we know, and how well

BEFORE THE DAY — TAG

Catch Whitlock before she drives off

Whitlock, the assistant operator, has the only relay settings that were never filed, and she is walking to her van. Once she is off site they are a week away. The whole afternoon rests on a number in one person's notebook.

PLAN CARD 71W

Friday. The fortnight goes into a report, and Dolores Reyes will not sign one that states everything at the same confidence. A reader who cannot tell a measurement from a guess will act on both the same way. June Farrow has the evidence file. Ravi Lindgren wants his forecast written as a range. Today you sort the fortnight's claims by how well each is known, and decide what becomes permanent practice.

BRIEFING 18W

The event report has to state what is established, what is inferred and what is still a guess.

WHAT YOU DECIDE 15W

Say how strongly each of the fortnight's claims is held, and what outlives the emergency.

10.1 Measured, worked out, or suspected

METERING & STANDARDS · A ROOM · ATTEST

SCENE 27W

June Farrow has four statements ready for the final report and two checks left before it is signed. Some are already backed. Others are still a suspicion.

GUIDE 57W

Four statements are ready for the report and you have two checks. Open each claim and read what already backs it. Some are measured, some are worked out from something measured, and some are only suspected. Hold the ones the backing does not support, and spend the checks where a wrong claim would do the most damage.

ASK 10W

Which claims deserve the two checks before they become findings?

BEHIND THE BUTTON 132W

The three kinds, and why they look the same. A measured claim points at a record. A worked-out claim is a chain of reasoning from a record, and it is only as good as its weakest step. A suspicion has neither. In a report all three are just sentences.

Why damage decides the spend, not doubt. The weakest claim is not automatically the one to check. A shaky statement nobody will act on costs little. A confident one that will send somebody to replace a piece of equipment costs a lot if it is wrong.

What holding a claim means. Not deleting it. Marking it as not yet supported, so the report says what it can stand behind. A finding is something the author is willing to be wrong about in public.

TAKES AS READ

a claim can be measured, worked out, or still only suspected · there is not enough time to check everything

VERDICT 79W

The four claims are not equally supported. The sensor being wrong has been shown two separate ways, with a date to match, so it is settled. The wire having lost strength is good reasoning and nobody has tested a piece of it. The store's efficiency is measured, and it is not what the inquiry turns on. The relay explanation is both central and unproved. Two checks, and they belong on the claims that matter and have nothing under them.

TAKEAWAY 17W

A signature is only worth the evidence behind it, so check the important claims that have none.

10.2 The number that has to be a range

LOAD & FORECASTING · A PERSON · CHOICE

SCENE 25W

Lindgren's winter peak goes into the report. He wants the width written into the sentence, and Reyes wants to know what the width is for.

GUIDE 59W

Four reasons for a range, and they differ in what the width is for. Ask of each whether it is about blame, measuring, or planning. Reserve is not held for the expected peak. It is held for how far above it the peak might land, and that is the width. One figure deletes the only thing a planner needs.

ASK 14W

Why does the winter peak go in as a range rather than one figure?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a forecast covers a spread of outcomes, not just one

VERDICT 80W

Reserve is not held against the peak everybody expects. It is held against how far above that the peak might land, which is a fact about the width and not about the middle. Print a single number and the one quantity a planner needs has been deleted, so the planner invents one, usually too small. The width is also the honest part of the forecast. It says which conditions the model was built on and which it has never seen.

TAKEAWAY 16W

A prediction is only usable by a planner if it says how wrong it might be.

10.3 What outlives the emergency

SYSTEM OPERATIONS · A ROOM · CHOICE

SCENE 26W

Four things the room started doing under pressure. Reyes asks which are worth their cost on an ordinary Monday, when nothing at all is going wrong.

GUIDE 58W

All four changes are fair, and all four cost something every day. Ask of each whether it would have caught this fortnight's mistake. Then ask whether the room will still be doing it in March. The sensor was wrong for months because nothing was ever compared with it. More reports of a wrong number are not more information.

ASK 9W

Which of the fortnight's changes is most worth keeping?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a habit taken up in an emergency has a cost that only shows later

VERDICT 61W

One of these would have caught the whole thing. The sensor was wrong for months, because nothing was ever compared against it. Comparing it costs an afternoon. Extra reserve costs money every night for a benefit on a few of them. Reading the loading four times as often would have changed nothing at all. Those readings were wrong, rather than late.

TAKEAWAY 17W

Keep what would have caught it early. Drop what only paid off because the fortnight was strange.

10.4 What the desk is chasing now

DISTRIBUTION · DISTRIBUTION DEPOT · A ROOM · SPOT

SCENE 21W

Faults arrive on the board all evening. The desk works to one priority, and it is changed as the picture changes.

GUIDE 37W

Take the faults the priority asks for and leave the rest. The priority at the top of the board changes during the evening and nobody announces it. What counts is the faults either side of a change.

ASK 13W

Send crews to the priority on the board, and keep checking the board.

BEHIND THE BUTTON 65W

Why the priority changes. Early on it is getting the most customers back. Then a hospital goes onto its generator, and it becomes that hospital. Later it is whatever cannot be left out overnight in bad weather.

Why the change is where it goes wrong. A desk still using the old priority sends its last free crew to the job that mattered an hour ago.

TAKES AS READ

a control desk works to a priority somebody can change

VERDICT 110W

Three priorities run across the evening and each wants different faults. Most customers back first, then anything with a vulnerable customer behind it, then anything that cannot be left out overnight in the weather. A fault can match two at once, and that is what makes a change cost something. What the panel scores is the faults either side of a change. Everything else on the board is wanted by neither priority, so ignoring those proves nothing. The desk's own version is the ten minutes after a hospital calls. The priority has changed, and the crews already driving are driving to the order that was in force when they left.

TAKEAWAY 18W

The cost of a rule that has changed is paid by whoever is still using the old one.

10.5 Three things that have to agree

GENERATION · GENERATION HALL · A ROOM · CALLBACK · SEQUENCE

SCENE 30W

Haddad has a machine at rest and a grid running normally. She wants the order written down before anybody is standing at the panel with a hand on the switch.

GUIDE 61W

All three of these happen at the panel, so nothing is missing from the list. Ask of each what difference it removes. Then ask what that difference would do if the switch closed while it was still there. A speed difference and a voltage difference break things in different ways. One of these three only works once the others are right.

ASK 12W

Put the steps of joining a machine to the grid in order.

BEHIND THE BUTTON 153W

Why the order is graded whole. A sequence is a claim that each step needs the one before it to have happened. Get one pair the wrong way round and the claim is wrong, even if everything else lines up. That is why there is no part credit.

Why the ends are the place to start. The first and last steps are usually the easiest to be sure of, because one has nothing before it and the other has nothing after it. Fixing those two leaves fewer ways for the middle to go, and the middle is where the real question usually is.

What the rail can be ordered on. Often it is time. It can also be cost, or risk, or what you can still undo afterwards — do the steps that change nothing before the ones that change something for good. A step that destroys the sample can only be last.

TAKES AS READ

a generator joined to a live grid is forced to turn at the grid's speed

VERDICT 74W

Closing that switch ties the machine to the whole grid, and the grid does not adjust. So every difference has to go first. Speed first, because a machine turning at the wrong speed gets slammed into step, hard enough to damage it. Voltage next, or current pours between them for no useful reason. Then wait for the two waves to line up. Close them when all three match, and the machine simply joins in.

TAKEAWAY 16W

Every mismatch gets removed before the switch closes, and there is only one order that works.

END OF THE DAY 21W

A finding is worth what its evidence is worth, and saying so is what makes a report usable a year later.

THE CLOSING CARD

The valley was back inside three hours and the report went out on the Friday, with the findings ranked by the evidence under them and the cause of the trip written down as an open question. The corridor sensor reads true now. The second circuit is strung and energising next autumn, which means eighteen more months of the corridor you have been holding all fortnight, at the strength you now know it has.

What it cost: seven hundred and fifty megawatt-hours never delivered, a fifth of an island shed to hold its frequency, and ninety hours of a conductor quietly losing strength that nobody had added up. What is unfinished: the trip has no proven cause, the store has been measured once, and the practice that would have caught all of it — check any instrument a decision rests on against something independent — is a paragraph in a report until somebody on nights makes it a habit.

Card text only — no options, boards or answers. 7,858 words of card prose across 41 stops. Generated from `themes/blackout_ms` by `tools/make-cardbook.mjs`.